

Java Assignment 4

Q1: Online Shopping System

Design a system for an e-commerce platform to manage products, orders, and payments.

1.1 Create a class Product with:

- productId (int)
- productName (String)
- price (**String** input from user)

1.2 Add a method in Product to convert price from String to double using **Wrapper class**.

1.3 Create a class Order with:

- customerName (String)
- array of Product

1.4 Write a method in Order to:

- iterate through all products
- fetch numeric price
- calculate and return total bill

1.5 Overload the total calculation method:

- one method → normal total
- second method → total after applying discount (%)

1.6 Create a method to generate bill using:

- StringBuilder or StringBuffer
- Include:
 - customer name
 - list of products
 - total amount

1.7 Create an interface Payment with:

- pay(double amount)
- refund(double amount)

1.8 Create two classes:

- UPIPayment
- CardPayment

Implement Payment and provide different messages (**method overriding**).

1.9 In main method:

- take user input
- create products and order
- ask user to choose payment type
- call appropriate payment method

1.10 Add **exception handling** for:

- invalid price (NumberFormatException)
- product count ≤ 0
- invalid payment option

1.11 Display:

- formatted bill
- final payable amount
- payment success message

Q2: Smart Banking System

Design a system to manage bank accounts and transactions.

2.1 Create a base class Account with:

- accountNumber (int)
- accountHolderName (String)
- balance (double)

2.2 Create derived classes:

- SavingsAccount
- CurrentAccount

using **inheritance**.

2.3 Override a method displayAccountDetails() in both child classes to show account type (**method overriding**).

2.4 Create an interface BankService with:

- deposit(double amount)
- withdraw(double amount)

2.5 Implement BankService in account classes and define logic for deposit and withdrawal.

2.6 Accept balance as **String input** and convert it into double using **Wrapper classes**.

2.7 Overload deposit method:

- deposit(amount)
- deposit(amount, bonus)

2.8 Implement withdrawal method:

- subtract amount from balance
- ensure balance does not go below zero

2.9 Create a method to generate account summary using:

- StringBuilder or StringBuffer
- Include:
 - account details
 - current balance

2.10 Add **exception handling** for:

- invalid numeric input
- negative deposit/withdrawal
- withdrawal greater than balance

2.11 In main method:

- take input
- create account object
- perform deposit and withdrawal
- display updated account summary